MIL-M-28787/227A(EC)
28 November 1978
SUPERSEDING
MIL-M-28787/227(EC)
13 May 1977

MILITARY SPECIFICATION
MODULES, STANDARD ELECTRONIC
TRANSFORMER, RESOLVER, ANALOG, KEY CODE PTF

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic resolver transformer module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Name	Transformer, resolver
Family	Analog
Key code	PTF
Size	1A
Weight	35 grams maximum
Failure rate	0.032 failures/10^6 hours maximum
Environment	Class I as specified in MIL-STD-1389
Individual transformer description:	
Frequency	390 to 820 Hz
Dielectric withstanding	200 ±15 Vdc
Transformer ratio	0.29661 ±0.00296 (11.8 : 3.5) 0.08475 ±0.00085 (11.8 : 1.0)
Regulation	0.1% from no load to 50 kΩ resistive load
Phase shift	0° to 0.1° (50 kΩ load)
Input impedance (V_I = 11.8 Vrms)	70 kΩ minimum at 400 Hz 100 kΩ minimum at 800 Hz
Input voltage	11.8 Vrms ±10%

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Individual transformer description:
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Angular error	0.008° maximum
Clamping circuit voltage	±10 Vpk
Quadrature - Full winding	±1.0 mV
Tap	±0.5 mV

1.3 Module function. The resolver transformer module, designated as key code PTF, is a Standard Electronic Modules Program (SEMP) electronic module with 0.100 inch (2.54 mm) grid center contacts. This module contains two 390 to 820 Hz transformers which convert 11.8 Vrms resolver signals to 3.5 and 1.0 Vrms resolver signals. Each secondary circuit has two suppression circuits to limit output transients to ±10 Vpk.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(All military standards and military specifications shall be ordered from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

3. REQUIREMENTS

3.1 General requirements The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.032 failures per million operation hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

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3.5 Module configuration requirement. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be PTF.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 20 contact pins positioned A1 through A20.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/227, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The functional diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table I.

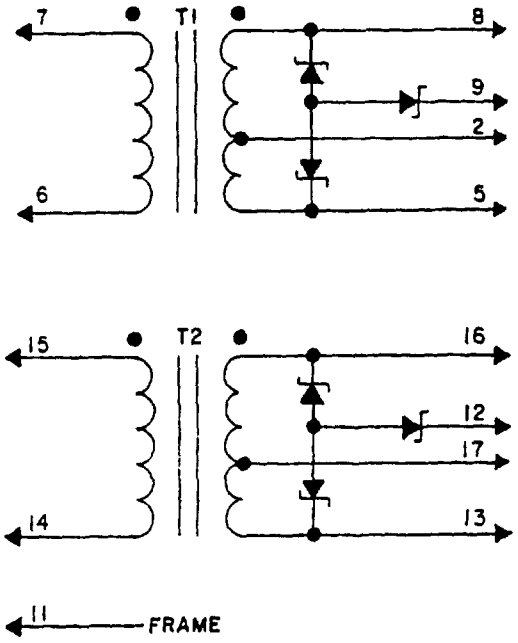
3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

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NOTES.
1. Pins 1, 3, 4, 10, 18, 19, and 20 are unused.
2. All diodes shall be JAN TX 1N4465, or equivalent

FIGURE 1. PTF module functional diagram, resolver transformer.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ± 5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A4	Isolation	II	X	10	-	1.0	-	10	-	1.0	-	MΩ
B	Frame continuity	III	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ω
C1, C2	DC resistance	IV	S	70	95	70	100	60	120	60	120	Ω
C3, C4	DC resistance	IV	S	25	35	25	35	20	45	20	45	Ω
D1, D2	Suppression voltage	V	Q	16.0	26.0	16.0	26.0	16.0	26.0	16.0	26.0	Vp-p
D3-D6	Suppression voltage	V	X	16.0	26.0	16.0	26.0	16.0	26.0	16.0	26.0	Vp-p
E	Polarity	VI	X	3.2	3.8	3.2	3.8	3.2	3.8	3.2	3.8	V
F1-F3, F8	Angular accuracy check	VII	X	-0.004	+0.004	-0.004	+0.004	-0.004	+0.004	-0.004	+0.004	Degree
F4-F7	Angular accuracy check	VII	X	-0.008	+0.008	-0.008	+0.008	-0.008	+0.008	-0.008	+0.008	Degree
F9-F12, F16-F19, F21-F24	Angular accuracy check	VII	S	-0.008	+0.008	-0.008	+0.008	-0.008	+0.008	-0.008	+0.008	Degree
F13-F15 F20	Angular accuracy check	VII	S	-0.004	+0.004	-0.004	+0.004	-0.004	+0.004	-0.004	+0.004	Degree
G1-G7, G13-G19	Differential quadrature	VIII	S	-1.0	+1.0	-1.0	+1.0	-1.0	+1.0	-1.0	+1.0	mV
G8-G12 G20-G24	Differential quadrature	VIII	S	-0.5	+0.5	-0.5	+0.5	-0.5	+0.5	-0.5	+0.5	mV
H1, H2	Common mode rejection ratio check	IX	Q	-74	--	-74	--	-74	--	-74	--	dB
H3, H4	Common mode rejection ratio check	IX	Q	-60	--	-60	--	-60	--	-60	--	dB

See notes at end of table.

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TABLE I. Module electrical requirements -Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ± 5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
H5, H6	Common mode rejection ratio check	IX	Q	-40	--	-40	--	-40	--	-40	--	dB
J1	Transformation ratio	X	Q	0.2936	0.2996	0.2936	0.2996	0.2936	0.2996	0.2936	0.2996	V/V
J2	Transformation ratio	X	Q	0.0839	0.0856	0.0839	0.0856	0.0839	0.0856	0.0839	0.0856	V/V
K1, K2	Phase shift check	XI	Q	-0.1	+0.1	-0.1	+0.1	-0.1	+0.1	-0.1	+0.1	Degree
L1, L2	Load regulation check	XII	Q	0.05	0.1	0.05	0.1	0.05	0.1	0.05	0.1	%
M1, M2	Input impedance check	XIII	X	80	250	75	400	75	350	70	500	kΩ
M3,M4	Input impedance angle check	XIII	Q	18	60	18	60	18	60	18	60	Degree
M5	Input impedance balance	XIII	X	0	20	---	---	---	---	---	---	%
M6, M7	Input impedance check	XIII	S	130	470	125	490	110	500	100	550	kΩ
M8, M9	Input impedance angle check	XIII	Q	5	45	5	45	5	45	5	45	Degree

NOTES:
1. (X) parameters shall be 100 percent tested during module acceptance tests.
2. (S) parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3. (Q) parameters shall be tested during qualification testing only.

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4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
Sine wave generator	Item G01	1
Power amplifier	Item A01	1
AC voltmeter	Item W01	1
Megohm bridge	Item R06	1
Ohmmeter	Item R01	1
Ohmmeter	Item R07	1
Oscilloscope	Item D01	1
Oscilloscope probe	Item Y04	1
AC voltage standard	Item G07	1
Synchro/resolver standard decade	Item F03	1
Phase angle voltmeter	Item Z01	1
Resolver/synchro bridge	Item F04	1
Frequency meter	Item T02	1
Ratio transformer	Item F02	2
Phase angle voltmeter	Item Z02	1
Resistors		
51Ω ±5%, 1/2 W		1
50 kΩ ±1%, 1/8 W		2
10Ω ±1%, 2 W		1
301Ω ±1%, 1/8 W		1
12.1Ω ±1%, 1/8 W		2
100Ω ±1%, 1/8 W		1
Switch		
SPDT		1

All resistors shall be metal film unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the SEMP Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through M shall be performed in any order with the following exceptions. If test condition A is part of a sequence of tests, the order of testing shall be as follows: A test, where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through M in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through M shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

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4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be nonoperating. Following the vibration tests, the electrical tests of table I which are marked with an (X) in the test type column shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test, and while still at the maximum class temperature, electrical tests C through M shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through M shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

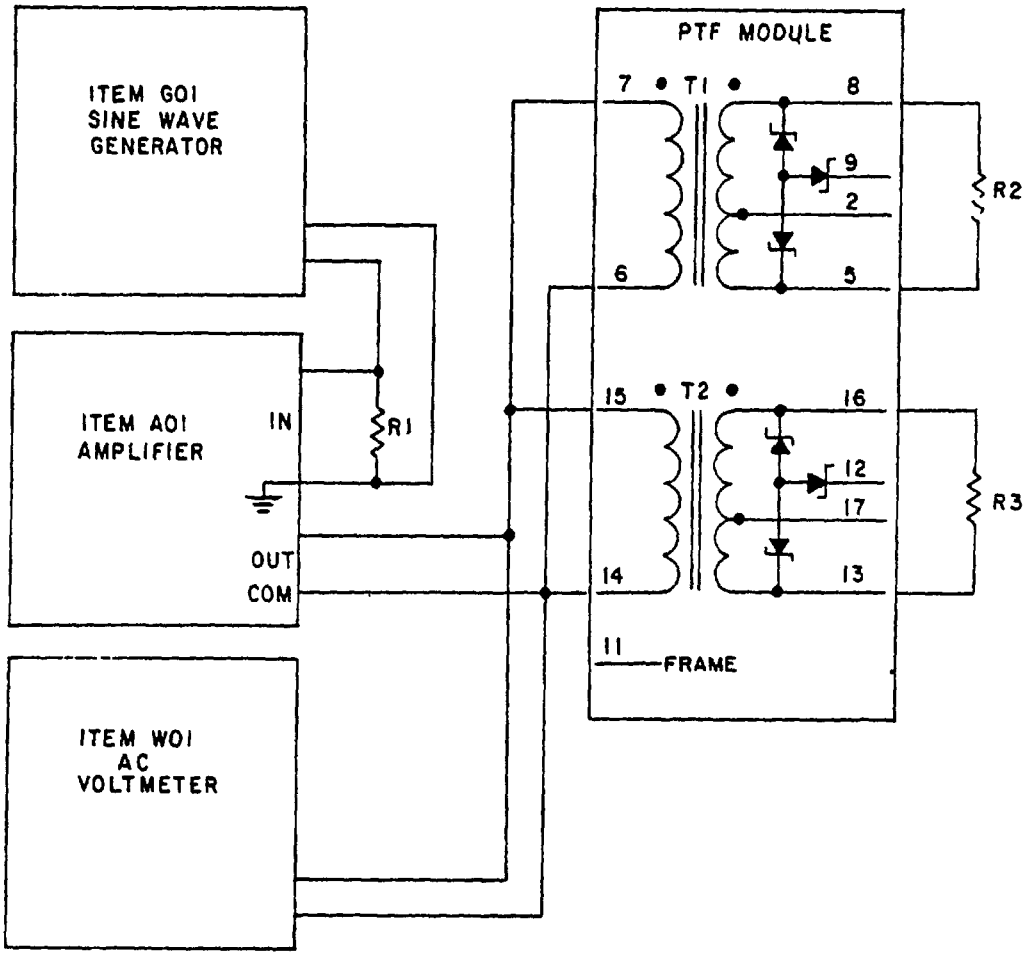
6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life.

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project MISC-NC71-45)

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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 = R3 = 50 kΩ ±1%, 1/8 W.
3. Adjust items A01 and G01 for an output voltage of 11.8 ±0.02 Vrms at 430 ±40 Hz.

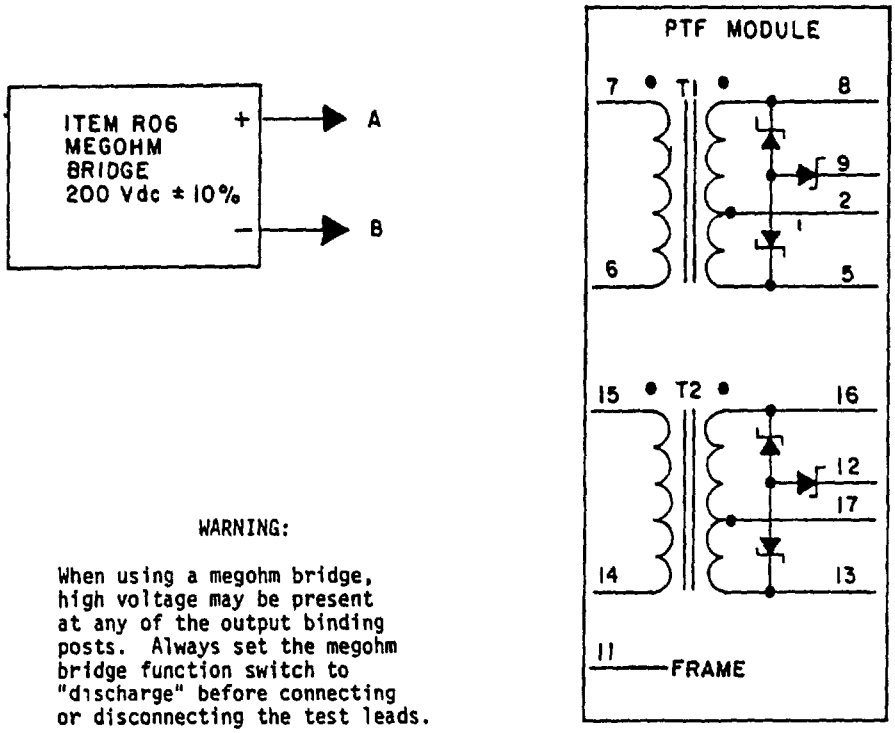
FIGURE 2. Operating temperature, component temperature, and life test circuit.

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	1,3,4,11,18	2,6,10,14, 17,19,20	10	∞	MΩ	3
A2	↓	2,6,11,14,17	1,3,4,10,18, 19,20	↓	↓	↓	↓
A3	↓	1,2,3,6,10, 19	4,14,17,18, 20	↓	↓	↓	↓
A4	Isolation	1,2,4,10,14	3,6,17,18,19	10	∞	MΩ	3

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

FIGURE 3. Isolation test circuit for table II.

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TABLE III. Frame continuity test.

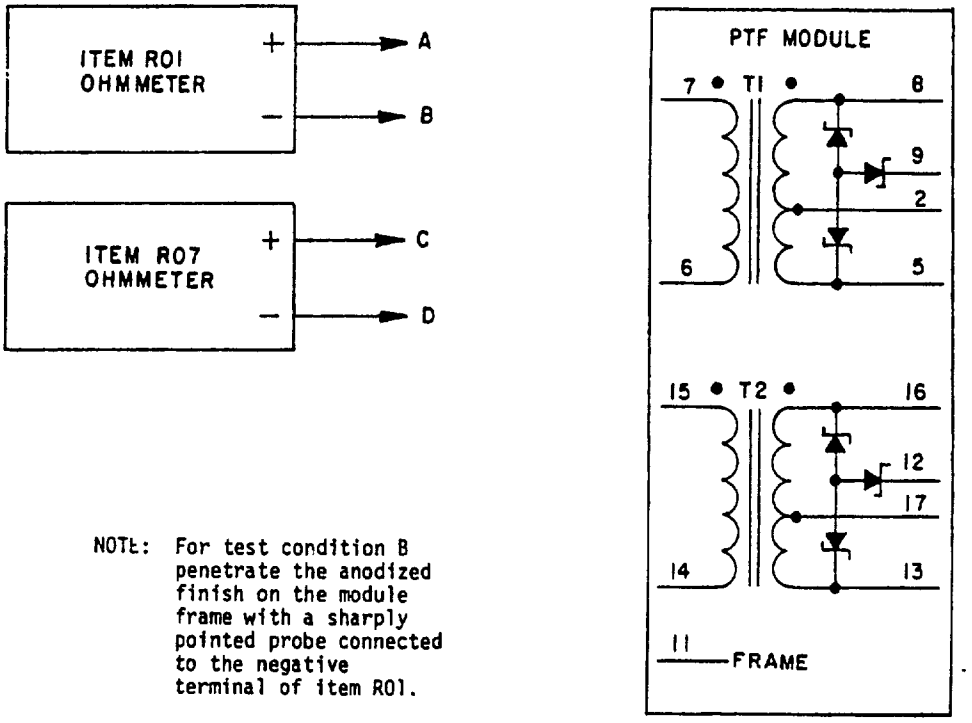
Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
B	Frame continuity	11	Frame	Open	Open	0.0	1.0	Ω	4

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TABLE IV. DC resistance test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
C1	DC resistance	Open	Open	6	7	70	95	Ω	4
C2	→	→	→	14	15	70	95	→	→
C3	→	→	→	5	8	25	35	→	→
C4	DC resistance	Open	Open	13	16	25	35	Ω	4

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NOTE: For test condition B penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of item R01.

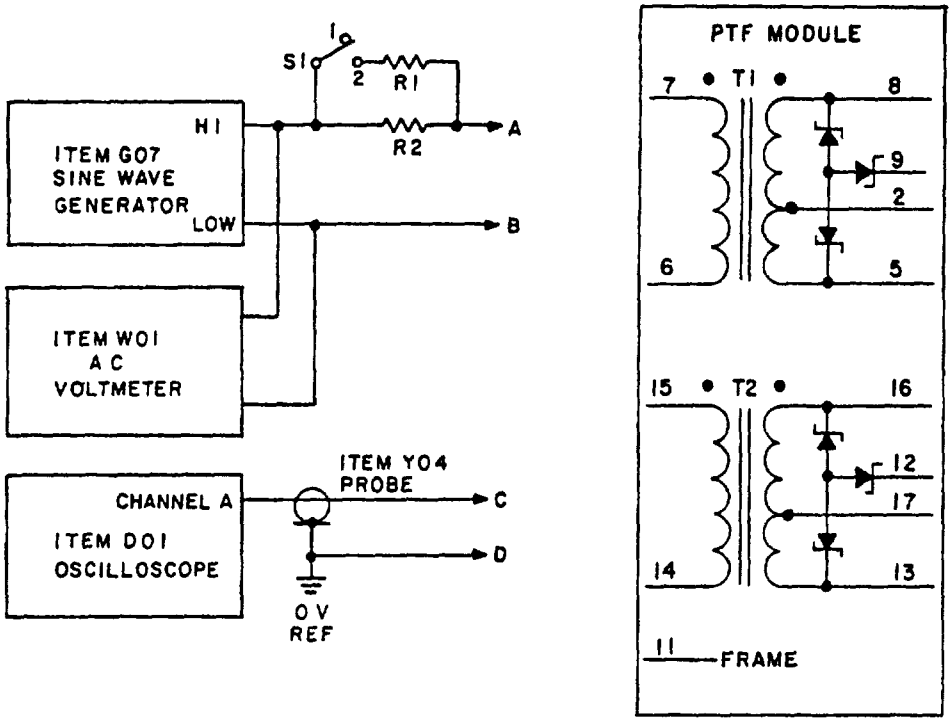
FIGURE 4. Frame continuity and dc resistance test circuit for tables III and IV.

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TABLE V. Suppression voltage test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D1	Suppression voltage	5,8	9	5,8	9	16.0	26.0	Vp-p	5
D2		13,16	12	13,16	12				
D3		5,8	9	5,8	9				
D4		13,16	12	13,16	12				
D5		6	7	8	5				
D6	Suppression voltage	14	15	16	13	16.0	26.0	Vp-p	5

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NOTES:
1. R1 = 10Ω ±1%, 2 W.
2. R2 = 301Ω ±1%, 1/8 W.
3. S1 = SPDT switch.
4. Test conditions D1 and D2:

   a. Connect the equipment as specified in table V for the appropriate test condition. Set S1 to position 2.
   b. Adjust item G07 for 10.9 ±0.2 Vrms at 2.0 ±0.2 kHz at the output.
   c. Measure the peak-to-peak voltage displayed on item D01.
   d. The measured voltage shall satisfy the limits of table V.

5. Test conditions D3 and D4:

   a. Connect the equipment as specified in table V for the appropriate test condition Set S1 to position 1.
   b. Perform test per notes 4b, 4c, and 4d.

6. Test conditions D5 and D6:

   a. Connect the equipment as specified in table V for the appropriate test condition. Set S1 to position 1.
   b. Adjust item G07 for 30.0 ±0.3 Vrms at 2.0 ±0.2 kHz at the output.
   c. Perform test per notes 4c and 4d.

FIGURE 5. Suppression voltage test circuit for table V.

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TABLE VI. Polarity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E	Polarity	8	16	3.2	3.8	V	6

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TABLE VII. Angular accuracy check.

Test condition	Parameter	Connect		Set F03 to angle (degree)	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin		Min (see note)	Max (see note)		
F1	Angular accuracy check	8	16	0.000	359.996	0.004	Degree	6
F2				45.000	44.996	45.004		
F3				90.000	89.996	90.004		
F4				20.000	19.992	20.008		
F5				70.000	69.992	70.008		
F6				110.000	109.992	110.008		
F7		8	16	160.000	159.992	160.008		
F8		2	17	45.000	44.996	45.004		
F9				20.000	19.992	20.008		
F10				70.000	69.992	70.008		
F11				110.000	109.992	110.008		
F12		2	17	160.000	159.992	160.008		
F13		8	16	0.000	359.996	0.004		
F14				45.000	44.996	45.004		
F15				90.000	89.996	90.004		
F16				20.000	19.992	20.008		
F17				70.000	69.992	70.008		
F18	Angular accruacy check	8	16	110.000	109.992	110.008	Degree	6

See note at end of table.

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TABLE VII. Angular accuracy check -Continued.

Test condition	Parameter	Connect		Set F03 to angle (degree)	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin		Min (see note)	Max (see note)		
F19	Angular accuracy check	8	16	160.000	159.992	160.008	Degree	6
F20	↓	2	17	45.000	44.996	45.004	↓	↓
F21		↓	↓	20.000	19.992	20.008		
F22				70.000	69.992	70.008		
F23				110.000	109.992	110.008		
F24	Angular accuracy check	2	17	160.000	159.992	160.008	Degree	6

NOTE: Table VII limits (degree) = angle set on item F03 ± (table I limits).

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TABLE VIII. Differential quadrature test.

Test condition	Parameter	Connect		Set F03 to angle (degree)	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin		Min	Max		
G1	Differential quadrature	8	16	0.000	-1.0	1.0	mV	6
G2				45.000				
G3				90.000				
G4				20.000				
G5				70.000				
G6				110.000				
G7		8	16	160.000	-1.0	1.0		
G8		2	17	45.000	-0.5	0.5		
G9				20.000				
G10				70.000				
G11				110.000				
G12		2	17	160.000	-0.5	0.5		
G13		8	16	0.000	-1.0	1.0		
G14				45.000				
G15				90.000				
G16				20.000				
G17				70.000				
G18				110.000				
G19	Differential quadrature	8	16	160.000	-1.0	1.0	mV	6

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TABLE VIII. Differential quadrature test -Continued.

Test condition	Parameter	Connect		Set F03 to angle (degree)	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin		Min	Max		
G20	Differential quadrature	2	17	45.000	-0.5	0.5	mV	6
G21	↓	↓	↓	20.000	↓	↓	↓	↓
G22				70.000				
G23				110.000				
G24	Differential quadrature	2	17	160.000	-0.5	0.5	mV	6

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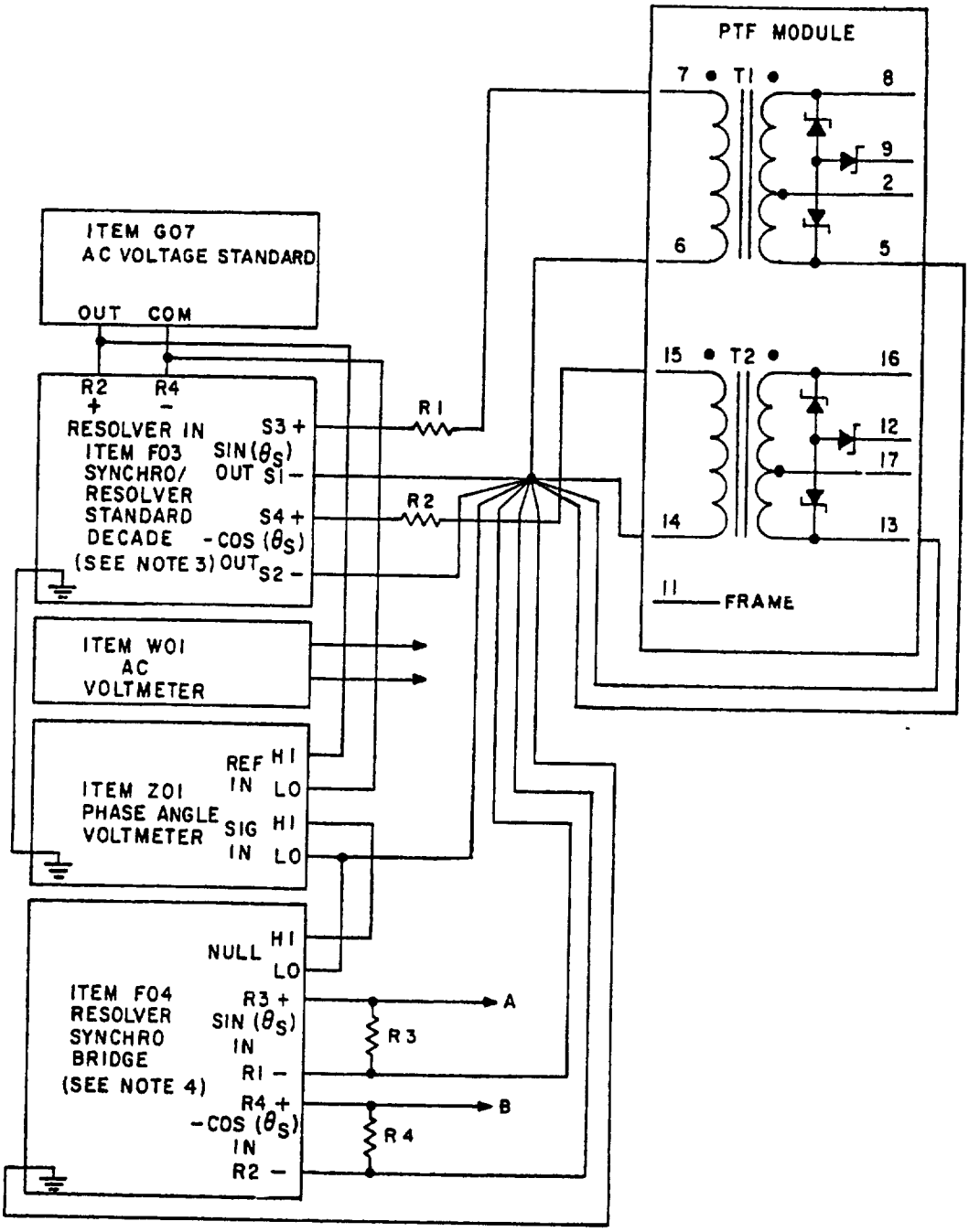


FIGURE 6 Polarity, angular accuracy check, and differential quadrature test circuit for tables VI, VII and VIII.

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NOTES:
1. R1 = R2 = 12.1Ω ±1%, 1/8 W.
2. R3 = R4 = 50 kΩ ±1%, 1/8 W.
3. For F03:
   $E_{(S3-S1)} = 0.454\ E_{(R2-R4)}\ SIN\ (\theta_S)$ and
   $E_{(S4-S2)} = -0.454\ E_{(R2-R4)}\ COS\ (\theta_S)$
   where (θ_S) = Standard angle.
4. For F04:
   $E_{(H1-LO)} = E_{(R3-R1)}\ COS\ (\theta_B) + E_{(R4-R2)}\ SIN\ (\theta_B)$
   where (θ_B) = Bridge angle.
5. Test condition E:

   a. Connect the equipment as specified in table VI.
   b. Adjust item G07 and item F03 for a 11.8 ±0.5 Vrms at 400 ±4 Hz resolver signal at the SIN θ resolver output of item F03 with the angle set at 90.000°.
   c. Set item F04 to 0.000°. Set item Z01 function switch to 0°. The voltage read on item Z01 shall satisfy the limits of table VI. The needle of item Z01 shall deflect to the right for an in-phase indication.

6. Test conditions F1-F12 and G1-G12 (performed concurrently):

   a. Connect the equipment as specified in table VII for the appropriate test condition.
   b. Repeat procedure of note 5b.
   c. With item F03 set to the angle specified for the appropriate test condition of table VII and item Z01 function switch set to 0°, adjust item F04 for a null reading on item Z01.
   d. The setting of item F04 shall satisfy the limits of table VII.
   e. With no change to the above test setup (6a-6d) set the function switch of item Z01 to 90°. The quadrature voltage read on item Z01 shall satisfy the limits of table VIII.

7. Test conditions F13-F24 and G13-G24 (performed concurrently).

   a. Perform test setup per note 6a.
   b. Adjust item G07 and item F03 for a 11.8 ±0.5 Vrms at 800 ±8 Hz resolver signal at the SIN θ resolver output of item F03 with the angle set at 90.000°.
   c. Perform tests per notes 6c, 6d, and 6e.

FIGURE 6. Polarity, angular accuracy check, and differential quadrature test circuit for tables VI, VII, and VIII - Continued.

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TABLE IX. Common mode rejection ratio check.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min (see note)	Max (see note)		
H1	Common mode rejection ratio check	7	6	8	5	0	2	mVp-p	7
H2	↓	15	14	16	13	↓	2	↓	↓
H3	↓	7	6	8	5	↓	10	↓	↓
H4	↓	15	14	16	13	↓	10	↓	↓
H5	↓	7	6	8	5	↓	100	↓	↓
H6	Common mode rejection ratio check	15	14	16	13	0	100	mVp-p	7

NOTE: Table IX limits (volts) = $10\left[10^{(x/20)}\right]$ where x = table I limits (dB).

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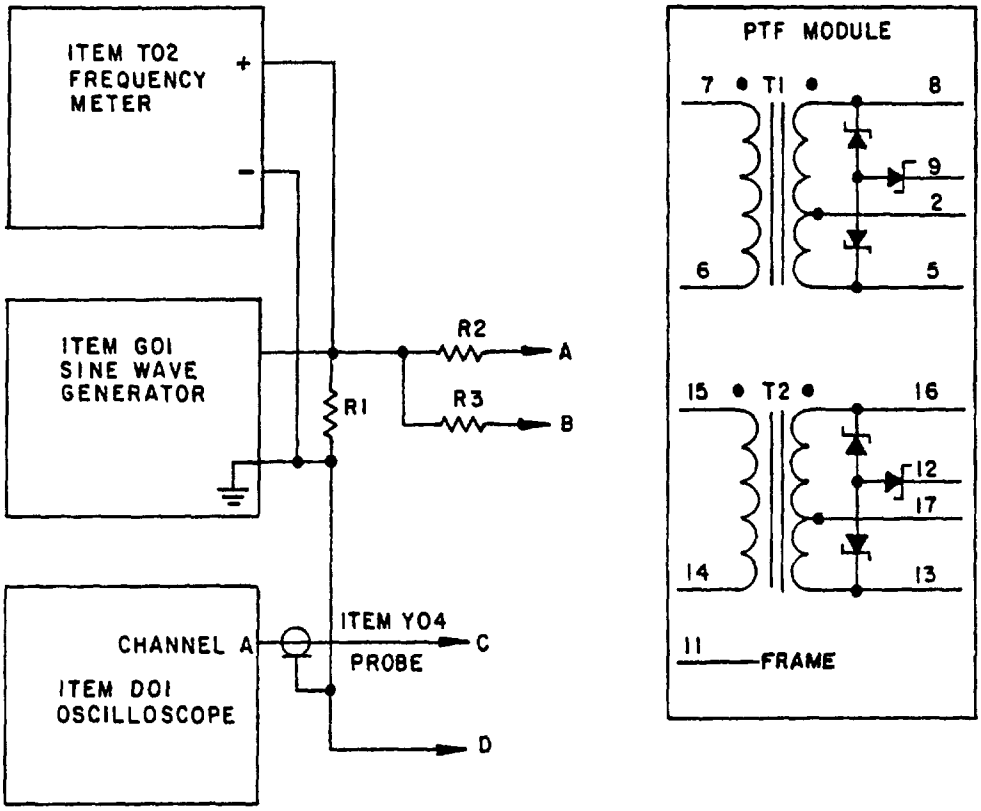


FIGURE 7. Common mode rejection ratio check test circuit for table IX.

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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 = R3 = 12.1Ω ±1%, 1/8 W.
3. Test conditions H1 and H2:

   a. Connect the equipment as specified in table IX for the appropriate test condition.
   b. Adjust item G01 for 10 ±0.1 V_{p-p} at 1.00 ±0.01 kHz at the output of item G01.
   c. The V_{p-p} measured on item D01 shall satisfy the limits of table IX.

4. Test conditions H3 and H4. Perform tests per notes 3a, 3b, and 3c with the frequency of item G01 set for 10.0 ±0.1 kHz.
5. Test conditions H5 and H6. Perform tests per notes 3a, 3b, and 3c with the frequency of item G01 set for 100 ±1 kHz.
6. The requirements (dB) of table I and the measured voltage of table IX have the following relationship:

$$dB = 20 \text{ LOG } \left[\frac{V \text{ (measured) (volts)}}{10 \quad \text{(volts)}} \right]$$

FIGURE 7. Common mode rejection ratio check test circuit for table IX - Continued.

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TABLE X. Transformation ratio test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J1	Transformation ratio	7	6	8	5	0.2936	0.2996	V/V	8
J2	Transformation ratio	7	6	2	5	0.0839	0.0856	V/V	8

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TABLE XI. Phase shift check.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min (see note)	Max (see note)		
K1	Phase shift check	7	6	8	5	-6	6	mV	8
K2	Phase shift check	7	6	8	5	-6	6	mV	8

NOTE: Table XI limits (volts) = 3.5 TAN [table I limits (degree)].

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TABLE XII. Load regulation check.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min (see note)	Max (see note)		
L1	Load regulation check	7	6	8	5	0.9990	0.9995	V/V	8
L2	Load regulation check	15	14	16	13	0.9990	0.9995	V/V	8

NOTE: Table XII limits = $1 - \left[\frac{\text{Table I limits (\%)}}{100} \right]$.

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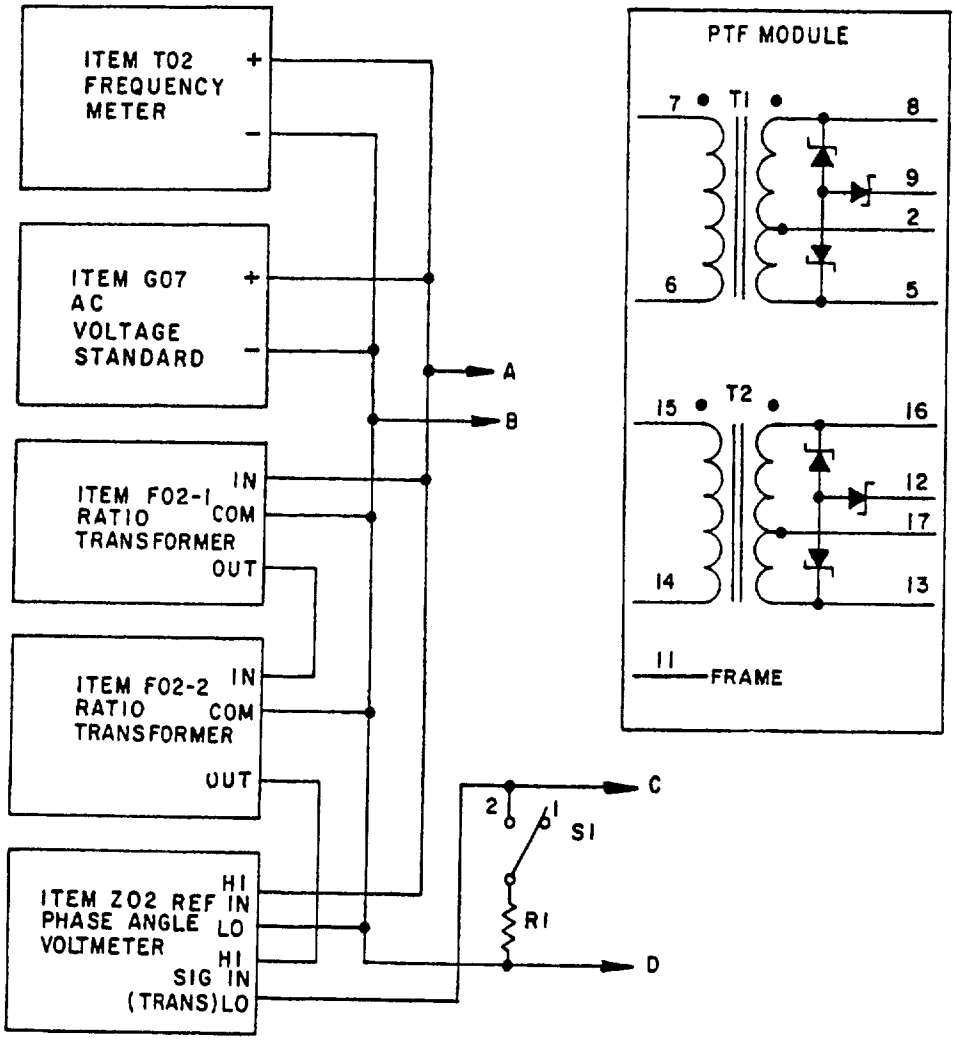


FIGURE 8. Transformation ratio, phase shift check, and load regulation check test circuit for tables X, XI, and XII.

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NOTES:
1. R1 = 50 kΩ ±1%, 1/8 W.
2. S1 = SPDT switch.
3. For test conditions J1, J2, K1, and K2, item F02-2 shall be set to 1.000000 and S1 shall be in position 1.
4. Test conditions J1 and J2:

   a. Connect the equipment as specified in table X for the appropriate test condition.
   b. Adjust item G07 for 11.8 ±0.2 Vrms at 400 ±4 Hz at the output.
   c. With the function switch of item Z02 set to 0°, adjust item F02-1 for a null reading on item Z02.
   d. The setting of item F02-1 shall satisfy the limits of table X.

5. Test condition K1:

   a. Connect the equipment as specified in table XI.
   b. Repeat procedure of notes 4b and 4c.
   c. With no change to the setup, set the function switch of item Z02 to 90° The quadrature voltage read on item Z02 shall satisfy the limits of table XI.

6. Test condition K2:

   a. Connect the equipment as specified in table XI.
   b. Adjust item G07 for 11.8 ±0.1 Vrms at 800 ±8 Hz at the output.
   c. Repeat procedure of note 4c.
   d. Switch the function switch of item Z02 to 90°. The quadrature voltage read on item Z02 shall satisfy the limits of table XI.

7. The requirements (degree) of table I and the voltage measured in test conditions K1 and K2 have the following relationship:

$$\text{Degree} = \text{ARCTAN}\left[\frac{\text{V (measured) (volts)}}{\text{3.5 (volts)}}\right]$$

8. Test conditions L1 and L2:

   a. Connect the equipment as specified in table XII for the appropriate test condition with S1 in position 1. Set item F02-2 to 1.000000.
   b. Repeat procedure of notes 4b and 4c.
   c. Set S1 to position 2 and adjust item F02-2 for a null reading on item Z02.
   d. The setting of item F02-2 shall satisfy the limits of table XII.

9. The requirements (%) of table I and the setting of item F02-2 have the following relationship:

   % = (100) (1-RT) where
   RT = Item F02-2 setting for the 0° null reading.

FIGURE 8. Transformation ratio, phase shift check, and load regulation check test circuit for tables X, XI, and XII - Continued.

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TABLE XIII. Input impedance check, input impedance angle check, input impedance balance test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
M1	Input impedance check	7	4.7 (note 1)	14.75 (note 1)	mV	9
M2	Input impedance check	15	4.7 (note 1)	14.75 (note 1)	mV	
M3	Input impedance angle check	7	30 (note 2)	72 (note 2)	Degree	
M4	Input impedance angle check	15	30 (note 2)	72 (note 2)	Degree	
M5	Input impedance balance	Open	0	20	%	
M6	Input impedance check	7	2.5 (note 1)	9.1 (note 1)	mV	
M7	Input impedance check	15	2.5 (note 1)	9.1 (note 1)	mV	
M8	Input impedance angle check	7	45 (note 2)	85 (note 2)	Degree	
M9	Input impedance angle check	15	45 (note 2)	85 (note 2)	Degree	9

NOTES:

1. Table XIII limits (volts) = $\frac{1{,}180 \text{ (volt-ohms)}}{\text{Table I limits (ohms)}}$

2. Table XIII limits (degree) = 90 - Table I limits (degree).

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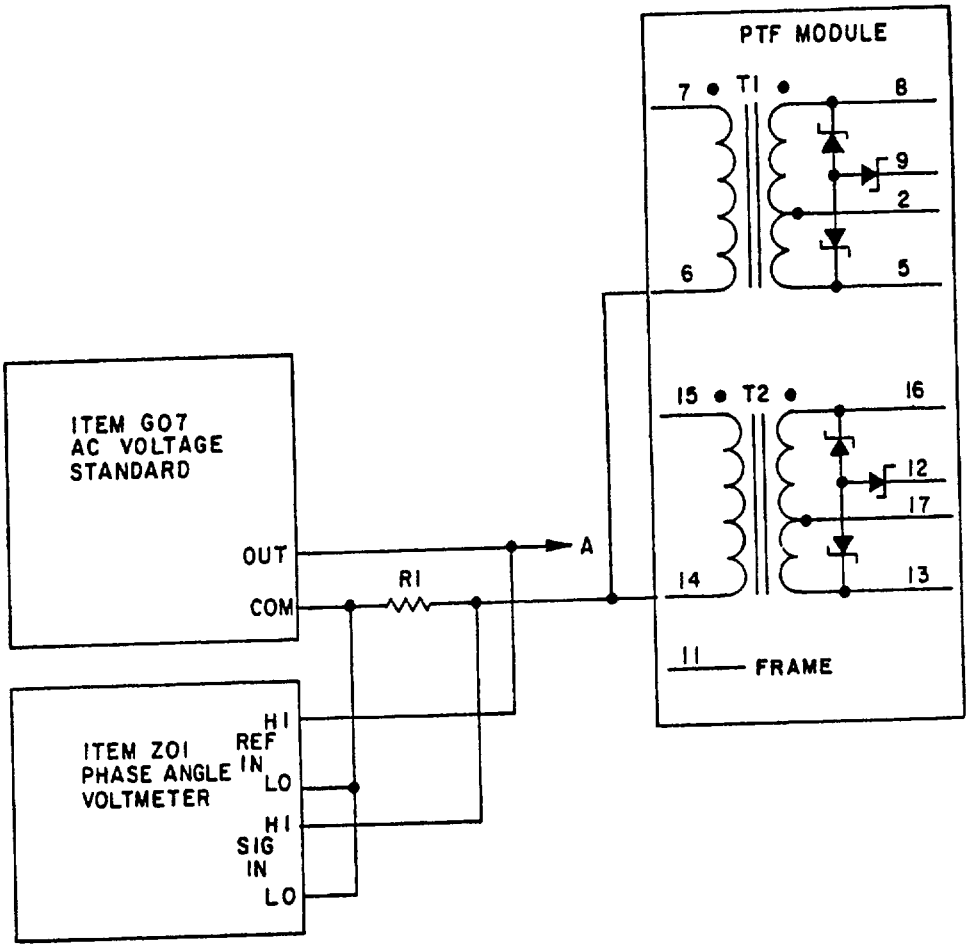


FIGURE 9. Input impedance check, input impedance angle check, and input impedance balance test circuit for table XIII.

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NOTES:
1. R1 = 100Ω ±1%, 1/8 W.
2. Test conditions M1 and M2:

   a. Connect the equipment as specified in table XIII for the appropriate test condition.
   b. Adjust item G07 for 11.8 ±0.2 Vrms at 400 ±4 Hz.
   c. With the function switch of item Z01 in the fundamental position, measure and record the voltage reading on item Z01. The reading shall satisfy the limits of table XIII.

3. Test conditions M3, M4, and M5.

   a. Repeat procedure of notes 2a and 2b.
   b. With the function switch of item Z01 in the 0° position, adjust the degrees dial for a null indication. The reading on the degrees dial shall satisfy the limits of table XIII. Set function switch to 90°, needle shall deflect to left.
   c. For test condition M5, requirement (%) of table XIII:

$$\% = \frac{A-B}{A} \times 100 \text{ where}$$

A = Larger voltage recorded for note 2c and
B = Smaller voltage recorded for note 2c.

4. Test conditions M6 and M7:

   a. Connect the equipment as specified in table XIII for the appropriate test condition.
   b. Adjust item G07 for 11.8 ±0.2 Vrms at 800 ±8 Hz.
   c. Perform tests per note 2c.

5. Test conditions M8 and M9:

   a. Repeat procedure of notes 4a and 4b.
   b. Perform tests per note 3b.

FIGURE 9. Input impedance check, input impedance angle check, and input impedance balance test circuit for table XIII - Continued.

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